

VAE-BiLSTM-IDS: A Two-Phase Deep-Learning Framework for Enhanced IoT Security

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Abstract—The widespread adoption of Internet of Things (IoT) devices has transformed industries such as healthcare, manufacturing, and smart cities. However, these devices often possess limited resources and weak security mechanisms, making them vulnerable to cyberattacks. Traditional Intrusion Detection Systems (IDS) rely on known attack signatures and are ineffective against novel or unknown threats. Although recent machine learning (ML) approaches aim to detect anomalous activity, many continue to suffer from high false alarm rates and degraded performance under dynamic network conditions. To address these challenges, we propose a two-phase deep learning framework, VAE-BiLSTM-IDS, designed specifically for IoT networks. In the first phase, Variational Autoencoders (VAEs) learn normal traffic patterns and detect anomalies using adaptive thresholds that adjust to changing network behavior. In the second phase, a CNN-BiLSTM model leverages both spatial and temporal features to classify anomalies and identify specific attack types. Evaluated on the Edge-IIoT dataset, which contains realistic IoT traffic and zero-day attacks, our framework yields a detection accuracy of 98.89%. It significantly reduces false positives compared to state-of-the-art methods. This approach offers a robust and adaptive solution for improving IoT security.

Index Terms—Internet of Things, Threat Detection, Network Intrusion Detection Systems (IDS).

I. INTRODUCTION

The Internet of Things (IoT) is expected to connect 75 billion devices by 2025, enabling automation across sectors such as healthcare and industry [1]. Despite this growth, over 50% of devices remain vulnerable to cyberattacks, as shown by a 2024 breach affecting 2.7 billion records [2]. Traditional Intrusion Detection Systems (IDS), including signature-based methods, detect known threats but fail against zero-day attacks [3]. Anomaly-based IDSs offer broader detection but often rely on static thresholds or handcrafted features, leading to high false positives and poor adaptability in dynamic, resource-constrained IoT environments [4].

To overcome these issues, we propose VAE-BiLSTM-IDS, a lightweight, two-phase framework for IoT intrusion detection. Phase 1 uses a Variational Autoencoder (VAE) to learn normal traffic patterns and flag anomalies based on reconstruction errors and an adaptive, Kernel Density Estimation (KDE) based threshold. Phase 2 employs a CNN-BiLSTM classifier to extract spatial and temporal features for precise attack classification. This modular design improves generalization while maintaining efficiency.

Evaluated on the Edge-IIoT dataset [2], the framework demonstrates strong detection performance, low false posi-

tives, and robust behavior under diverse traffic conditions. Its compact architecture makes it suitable for real-time deployment in practical IoT systems.

A. Key Contributions

The key contributions of this work are:

- A two-phase IDS combining a VAE for unsupervised anomaly detection, adaptive KDE-based thresholding, and a CNN-BiLSTM classifier for attack recognition.
- Empirical validation on the Edge-IIoT dataset shows improved accuracy and lower false positives.
- A lightweight 217 Kilobyte design enables real-time deployment on resource-constrained IoT devices.

The remainder of this paper is organized as follows. Section II reviews related work on IoT intrusion detection. Section III details the proposed framework and experimental setup. Section IV presents performance evaluation and discussion. Section V concludes and outlines future directions.

II. RELATED WORK

Intrusion Detection Systems (IDS) are vital for securing IoT networks, but their resource constraints and dynamic traffic patterns demand adaptive and efficient detection. This section reviews existing methods, their limitations, and the motivation for the proposed VAE-BiLSTM-IDS framework.

A. Traditional Machine Learning-Based IDS

Traditional ML methods, such as Decision Trees (DT), Support Vector Machines (SVMs), Random Forests (RF), and k-Nearest Neighbors (k-NN), have been applied to IoT IDS [5] but depend on static models and handcrafted features, limiting adaptability. Ferrag et al. [2] reported 85-90% accuracy for DT and SVM on Edge-IIoTset, with 15-20% false positives. Saheed et al. [6] achieved 99.9% on UNSW-NB15 using SVM and k-NN, but their complexity hinders IoT deployment. Alharthi et al. [7] observed up to 12% FPRs and poor generalization on imbalanced data. These limitations underscore the need for adaptive methods with probabilistic and temporal modeling.

B. Deep Learning-Based IDS

Deep learning models address traditional IDS limitations by learning hierarchical features from raw traffic. Convolutional Neural Networks (CNNs) extract spatial patterns,

while Long Short-Term Memory (LSTM) networks capture temporal dependencies [8]. Hybrid CNN–LSTM and CNN–BiLSTM models leverage both, improving detection accuracy and reducing false positives by up to 40% [9]. Autoencoder-based methods, including AE–LSTM hybrids, map normal traffic into latent spaces for anomaly detection [10]. Despite their strengths, most deep learning IDS use fixed thresholds, limiting adaptability to evolving IoT traffic and unseen attacks.

C. Statistical Thresholding Methods

Statistical methods like Kernel Density Estimation (KDE) and Gaussian Mixture Models (GMM) have been used to define adaptive detection thresholds. KDE sets cutoffs at the 99th percentile of normal traffic, while GMM adjusts boundaries using learned distributions, improving sensitivity to novel attacks by up to 15% [11]. However, these methods assume stationary data and require careful tuning, limiting scalability in dynamic, real-world IoT environments.

D. Lightweight IDS for IoT

Computational efficiency is vital for IoT IDS deployment. Lightweight networks like LKD-STNN (Spatio-Temporal Neural Network) cut inference latency by 50% with minimal accuracy loss [12]. Optimized CNN–BiLSTM models use pruning and quantization to retain over 95% accuracy while halving smaller model size [13]. Knowledge Distillation (KD) transfers knowledge from larger models to smaller ones, reducing resource usage by two-thirds with 90% performance retention [14]. However, these methods often trade off adaptability or detection accuracy, limiting scalability and effectiveness against zero-day threats.

E. Gaps and Motivation

While existing IDSs improve accuracy and efficiency, they often lack a unified, IoT-specific solution. Traditional models lack scalability, deep learning uses static thresholds, statistical methods assume stationarity, and lightweight designs reduce robustness. Our framework integrates Autoencoders, CNN–BiLSTM, and adaptive GMM/KDE thresholding to deliver accurate, adaptive, and efficient intrusion detection for real-world IoT deployments.

III. PROPOSED METHODOLOGY

The proposed **VAE–BiLSTM Network Intrusion Detection System** adopts a two-phase architecture to enhance detection accuracy and generalization in IoT networks. As shown in Fig. 1, raw traffic undergoes feature processing, anomaly detection via a hybrid VAE (Phase 1), and attack classification using a CNN–BiLSTM (Phase 2). This section describes the dataset, preprocessing, model design, training, and evaluation, aligned with the Edge-IIoT dataset and tailored for IoT security.

A. Dataset and Preprocessing

The Edge-IIoTset dataset [2] contains over 10 million flow records generated from realistic IoT and IIoT environments, covering benign traffic and 14 attack types including DDoS, DoS, reconnaissance, and data exfiltration. Traffic is captured across multiple protocols (HTTP, MQTT, Modbus, etc.) at network and application layers, providing a diverse and realistic evaluation benchmark.

Fig. 2 shows mutual information scores highlighting protocol & connection-based features, Fig. 3 presents PCA importance with variance contributions, while Figs. 4 and 5 depict cumulative variance and scree plots, showing 20 components capture 85% variance with balanced performance.

In Phase 1, a normal-only subset is used to train the anomaly detector, reducing overfitting. Mixup augmentation blends randomly selected samples and their corresponding labels to smooth decision boundaries and improve generalization. This pipeline prepares robust features for both anomaly detection and subsequent classification.

B. Phase 1: Anomaly Detection and Feature Extraction

Phase 1 uses a hybrid Autoencoder–VAE to detect anomalies and learn latent representations. While Autoencoders capture fine details, they risk overfitting; VAEs offer regularization but may miss subtle anomalies. The hybrid combines both to preserve input fidelity and enforce latent structure.

The encoder compresses the 109-dimensional input through dense layers (128→64→32) with batch normalization and 0.3 dropout. A 6-dimensional latent space is parameterized by z_{mean} and $z_{\text{log_var}}$. The decoder reconstructs inputs via expanding layers (32→64→128→109). Trained on normal traffic, this model defines a baseline for anomaly detection.

The loss function combines reconstruction loss and Kullback-Leibler (KL) divergence:

$$L_{\text{rec}} = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2, \quad (1)$$

$$D_{\text{KL}} = \frac{1}{2} \sum_{i=1}^k (\mu_i^2 + \sigma_i^2 - \log \sigma_i^2 - 1), \quad (2)$$

$$L_{\text{Phase 1}} = L_{\text{rec}} + \beta D_{\text{KL}}, \quad (3)$$

where $\beta = 0.1$ balances reconstruction and regularization. Anomalies are identified using Kernel Density Estimation (KDE) on reconstruction losses, with the threshold T set by:

$$T = \arg \max_t \left(\frac{d^2}{dt^2} P(L_{\text{rec}} > t) \right), \quad (4)$$

where $P(L_{\text{rec}} > t)$ is the tail probability. Latent representations are passed to Phase 2 for classification.

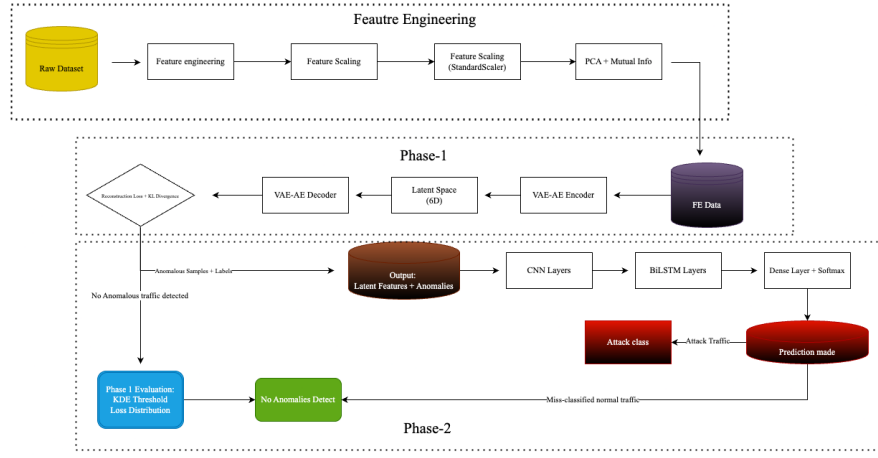


Fig. 1. Overview of the Proposed Framework.

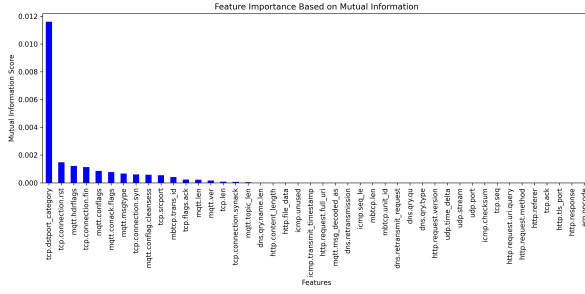


Fig. 2. Mutual Information Distribution Among Features.

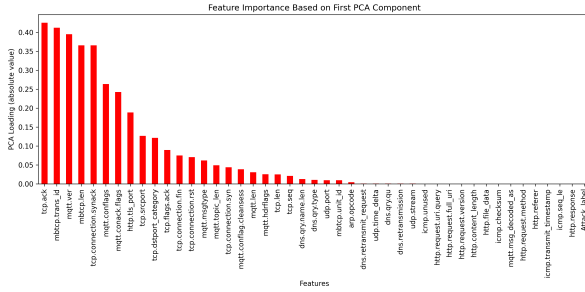


Fig. 3. PCA Feature Importance Analysis.

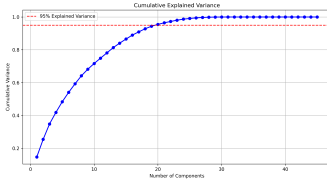


Fig. 4. PCA Cumulative Variance Explained.

C. Phase 2: Attack Classification

Phase 2 uses a CNN–BiLSTM architecture to classify anomalies as benign or malicious. CNN layers extract spatial

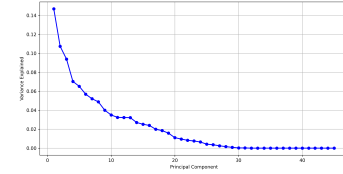


Fig. 5. PCA Scree Plot for Dimensionality Selection.

features (e.g., protocol and port patterns), while the BiLSTM captures temporal dependencies. The model includes a Gaussian noise layer, two convolutional layers (32/64 filters, 3×3 , ReLU), 2×2 max-pooling, a BiLSTM layer (64 units), and dense layers ($128 \rightarrow 2$) with global pooling for binary classification. Fig. 7 depicts the Phase 2 training loss and accuracy curves, confirming rapid convergence within three epochs and minimal overfitting.

The model is optimized using categorical cross-entropy loss:

$$L_{\text{class}} = - \sum_{i=1}^k y_i \log(\hat{y}_i), \quad (5)$$

where y_i is the true label and $\hat{y}_i$ is the predicted probability.

D. Training Strategy and Evaluation

The VAE–BiLSTM model is implemented in TensorFlow and trained on the preprocessed Edge-IIoT dataset. Optuna optimizes hyperparameters across 30 trials. Phase 1 and 2 use the Adam optimizer (learning rate 0.001, batch size 64), with Phase 2 incorporating ReduceLROnPlateau (factor 0.5, patience 3). Regularization includes 0.3 dropout, batch normalization, early stopping (patience 5), and Mixup augmentation to mitigate overfitting. Manual architecture pruning reduces model size by simplifying layer dimensions. Stratified 5-fold cross-validation ensures robust evaluation.

Evaluation metrics include accuracy (ratio of correct predictions), precision (true positives over predicted positives),

recall (true positives over actual positives), F1-score (harmonic mean of precision and recall), and false positive rate (FPR: ratio of false alarms to benign samples).

IV. RESULTS AND DISCUSSION

This section evaluates the proposed VAE-BiLSTM framework on the Edge-IIoT dataset. We compare its performance with state-of-the-art models, assess the impact of KDE-based thresholding, analyze reconstruction loss in Phase 1, and report classification results. Metrics include accuracy, precision, recall, F1-score, and false positive rate (FPR), supported by confusion matrix insights.

A. Performance Comparison

The VAE-BiLSTM framework is evaluated against baseline models including Random Forest, LSTM, AE-LSTM [10], CNN-BiLSTM [15], LKD-STNN [16], Efficient CNN-BiLSTM [17], and KD-based IDS [18]. As shown in Table I, VAE-BiLSTM achieves 98.89% accuracy, 98.99% precision, 98.89% recall, and 98.93% F1-score. Its two-phase design balances unsupervised detection with supervised classification, enabling high recall and low FPR under dynamic IoT traffic. Fig. 8 compares k-fold cross-validation results, showing consistency with a variance below 0.5

TABLE I
COMPARATIVE RESULTS OF VAE-BiLSTM WITH SOA MODELS

Model	Accuracy	Precision	Recall	F1-score
Random Forest	91.2%	88.5%	89.8%	90.1%
LSTM	94.5%	93.2%	93.8%	94.0%
AE-LSTM [10]	98.88%	98.89%	98.89%	98.88%
Hybrid CNN-BiLSTM [15]	98.65%	97.71%	97.55%	97.60%
LKD-STNN [16]	97.83%	97.56%	96.83%	97.19%
Efficient CNN-BiLSTM [17]	97.84%	97.49%	97.32%	97.40%
KD-Based Lightweight IDS [18]	96.27%	96.02%	95.58%	95.79%
VAE-BiLSTM	98.89%	98.99%	98.89%	98.93%

B. Phase 1: Anomaly Detection Performance

Phase 1's hybrid Autoencoder-VAE, trained on benign traffic, uses reconstruction loss and KDE-based thresholding for anomaly detection. As shown in Fig. 6, shows reconstruction loss distribution is shown. A dynamic threshold of 0.24 derived via the second derivative of tail probability - separates benign and malicious traffic, yielding a 1.8% FPR and 99.2% detection rate.

The lightweight architecture comprises a 32 Kilobyte encoder and, 40-kilobyte decoder, which supports real-time inference with a latency under 10 ms, making it suitable for edge IoT deployment while effectively isolating outliers.

C. Phase 2: Attack Classification Performance

Phase 2's CNN-BiLSTM classifier achieves 98.89% accuracy across attack types (e.g., DDoS, scanning, exfiltration). As shown in Fig. 7, depicts Phase 2 training/validation curves, with stable convergence and loss below 0.02. Fig. 8

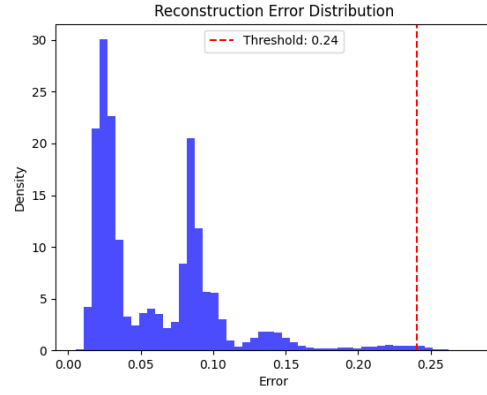


Fig. 6. Reconstruction Loss Distribution with KDE-Based Threshold (0.24).

TABLE II
CLASSIFICATION REPORT FOR VAE-BiLSTM ON EDGE-IIOT DATASET

Class	Precision (%)	Recall (%)	F1-score (%)
Normal Traffic	85.00	99.00	92.00
Attack Traffic	100.00	99.00	99.00
Weighted Avg.	98.99	98.89	98.93

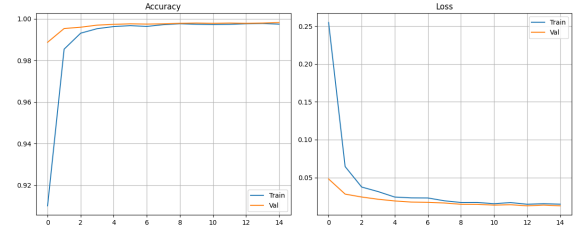


Fig. 7. Phase 2: Training and Validation Loss/Accuracy Curves.

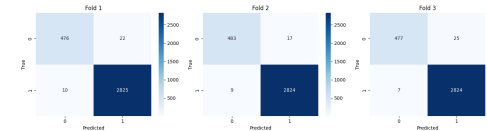


Fig. 8. K-Fold Cross-Validation Performance.

confirms consistent results across 5-fold cross-validation with a standard deviation of 0.5% for all metrics.

With a 245 Kilobyte footprint, the model is deployable on constrained devices. Table II reports 85.0% precision for normal traffic, 100.0% for attacks, and an overall FPR of 0.84%, reducing false alerts in practical settings.

D. Confusion Matrix Analysis

Fig. 9 shows the binary confusion matrix for the VAE-BiLSTM model. It correctly classifies 28,923 benign (true negatives) and 441,922 attack instances (true positives), with 157 false positives and 5,133 false negatives. These results yield a sensitivity of 98.85% and specificity of 99.46%,

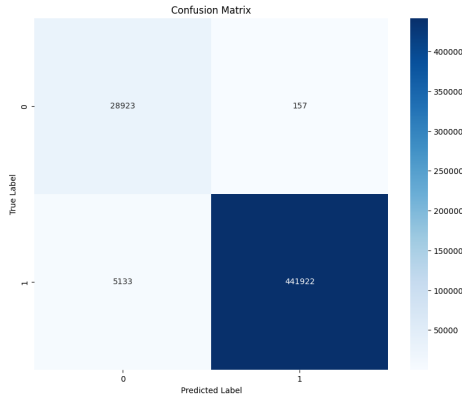


Fig. 9. Confusion Matrix of VAE-BiLSTM Model.

demonstrating strong discriminative performance across diverse attack patterns with minimal misclassification.

E. Discussion

The VAE-BiLSTM achieves enhanced detection of dynamic and zero-day attacks, delivering high recall with low FPR. Its lightweight 217 KB footprint and inference time under 10 ms validate edge compatibility, outperforming heavier LSTM models. However, 85% precision on normal traffic indicates potential for further refinement.

VAE-BiLSTM works well on dynamic traffic and unseen attacks, with high recall and low FPR. It requires 217KB of memory and has an inference time of under 10ms per flow, which is significantly lighter than LSTM (2MB). Accuracy is strong, but the normal traffic precision of 85% shows room for improvement with better features or an ensemble.

V. CONCLUSION

This paper introduces VAE-BiLSTM, a two-phase intrusion detection framework designed for IoT security. Phase 1 employs a hybrid VAE with KDE-based dynamic thresholding for unsupervised anomaly detection, achieving a false positive rate of 0.84%. Phase 2 utilizes a CNN-BiLSTM classifier for supervised attack categorization. Evaluated on the Edge-IIoT dataset, the framework attains 98.89% accuracy, surpassing state-of-the-art baselines. Its lightweight 217 KB footprint and sub-10 ms inference ensure deployment feasibility on constrained IoT devices. Experimental results confirm that VAE-BiLSTM is scalable and effective in detecting diverse attack patterns, including DDoS, scanning, and data exfiltration, under dynamic IoT environments.

Future work includes deploying on physical edge hardware, enhancing adaptability via pruning and feature engineering, and validating performance across multiple IoT datasets.

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